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Joby Aviation

Flight Research

California, United States

Application for Guidance, Navigation and Controls Engineer

Dear Joby Flight Research Hiring Team,

I am applying for the Guidance, Navigation and Controls Engineer position in Joby's Flight Research team. Your software-first approach to rapidly developing and flight-testing novel aircraft closely matches the way I have built my experience: combining aircraft dynamics and controls with C/C++ simulation, embedded flight software, real-time signal processing, autonomous UAV integration, and physical testing. I am particularly motivated by the opportunity to deliver well-engineered flight-controls software and use simulation, analysis, and testing to advance aircraft from stabilization toward basic autonomy.

In the Autonomous Robotics Group at the University of Luxembourg, I develop C++ and Python autonomy software for physical UAVs using ROS 2, PX4, embedded Linux, cameras, IMUs, and OpenVINS. My work links perception and state estimation with mapping, planning, and control, and I directly support controlled flight trials. I formulate test plans and evaluate integrated performance and robustness through simulation, laboratory, hardware, and flight testing. Working across software, controls, perception, and vehicle integration has made me comfortable taking ownership of implementation, interface troubleshooting, test analysis, and technical documentation.

As AOCS/ADCS Lead for the JamSail 3U CubeSat, I translated the concept of operations into requirements, control modes, architecture, interfaces, and a verification plan. I developed embedded C flight software and a C++/CMake Linux simulation of rigid-body attitude dynamics and control, then implemented an EKF, TRIAD attitude determination, and control-state logic. I also performed FFT-based sensor-signal analysis, selected filter cut-off frequencies, implemented digital low-pass filters, and verified their performance in simulation and on hardware. This work strengthened both my closed-loop GNC development and the disciplined software and testing practices required for real-time aerospace systems.

At Airbus Defence & Space, I developed C/C++ software on Linux for an offline GNC simulation that generated synthetic camera imagery and LiDAR scans for rapid algorithm development and performance analysis. As lead test engineer for LiDAR activities during Integrated BreadBoarding 3 and 4, I designed an integrated hardware/software test bench, formulated procedures, analyzed campaign data, and supported end-to-end verification at Technology Readiness Level 6. My First-Class MEng further developed my aircraft-dynamics foundation: I achieved 99% in Dynamics & Flight Mechanics and built MATLAB tools to analyze longitudinal and lateral motion and dynamic modes against Cranfield flight-test data.

Joby's small, agile Flight Research team is especially appealing because it requires engineers who can move fluently between dynamics, controls, software, simulation, analysis, and flight test. I would welcome the opportunity to bring that range, together with a strong sense of ownership and collaborative working, to the development and rapid iteration of high-quality flight controls for novel CTOL and VTOL aircraft.

Yours sincerely,

Jason Wakim Richards